IAT New Test 6

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

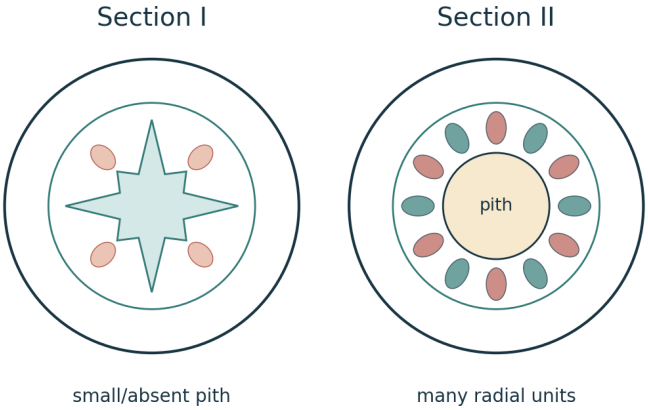
Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 Two organisms share a genus but belong to different species. A third is in a different genus of the same family. Which pair is guaranteed to share the most specific named rank?
A) First two: genus; all three: family
B) First two: family; all three: order
C) First two: species; all three: genus
D) All three: genus and species

Q2 A transverse root section in the figure has a central star-shaped xylem, phloem between its arms, and no conspicuous central pith. Another section has many xylem-phloem units around a large pith. Which assignment follows from both clues?



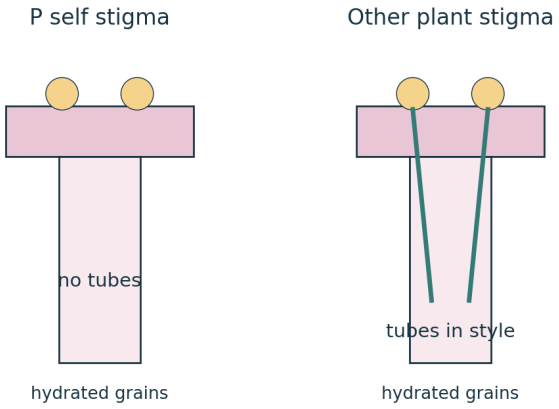
A) Both monocot roots at different ages
B) First monocot stem; second dicot stem
C) First dicot root; second monocot root
D) First dicot stem; second monocot root

Q3 In a translation experiment, a ribosome moves along an mRNA from its start codon. A mutation introduces a stop codon after 48 sense codons; the original stop followed 75 sense codons. How many fewer amino acids are incorporated, assuming no other change?
A) 26
B) 27
C) 28
D) 75

Q4 Isolated thylakoids are illuminated with electron transport operating, but a membrane-permeable proton carrier is added. Oxygen evolution continues briefly while ATP synthesis drops. Which direct change best explains the decoupling?
A) The carrier collapses the proton gradient across the thylakoid membrane
B) The carrier prevents water from donating electrons to PSII
C) The carrier removes the ADP substrate from the stroma
D) The carrier makes ATP synthase a source of photons

Q5 A patient's circulating thyroxine falls after reduced thyroid output, while the hypothalamus and pituitary remain functional. Which immediate feedback response is expected before treatment?
A) TRH falls and TSH rises
B) TRH and TSH both tend to rise
C) TRH rises and TSH falls
D) TRH and TSH both tend to fall

Q6 The diagram shows pollen from plant P placed on two stigmas: its own stigma and a genetically different stigma of the same species. Both grains hydrate, but tubes grow only in the latter style. Which mechanism first separates the outcomes?



A) Absence of viable pollen grains on either stigma
B) Failure of pollen dispersal from the anther
C) Self-incompatibility recognition after pollen deposition
D) Embryo abortion after double fertilisation

Q7 A replication fork advances rightward. The top parental strand is labelled 3′ at its left end and 5′ at its right end. Which new strand is synthesised continuously along that top template toward the fork?
A) No strand; the top template cannot be copied
B) A strand extending 3′ to 5′ toward the right
C) A strand extending 5′ to 3′ toward the left
D) A strand extending 5′ to 3′ toward the right

Q8 Before an antibiotic is applied, a bacterial culture contains a rare resistant variant. After several generations under the drug, most survivors are resistant. Which additional observation would directly reject the claim that the drug created the resistant mutation only after exposure?
A) The same resistance allele is detected in archived cells collected before exposure
B) The resistant population grows slowly without the drug
C) The drug inhibits a susceptible enzyme
D) The final population has fewer total cells

Q9 A restriction enzyme cuts a plasmid once and a DNA insert at both ends, yielding matching sticky ends. The enzyme is then replaced by one that makes blunt ends at all three cut sites; ligase is present. Which statement remains valid?
A) The insert must enter in exactly one orientation
B) The blunt-ended fragments cannot be joined by ligase
C) Ligation can occur, but matching sticky-end pairing no longer guides it
D) The original sticky overhangs remain after blunt cutting

Q10 A population grows logistically with carrying capacity K. At one census it has N=K/2; in an otherwise identical habitat another population has N=K. Which has greater instantaneous per-capita growth rate under the same intrinsic rate r?
A) The population at K
B) The population at K/2
C) Both are equal and positive
D) Both are zero

Q11 A bacterial operon reporter is expressed when an inducer is present. A strain lacking the repressor gene expresses the reporter even without inducer; restoring the repressor gene restores inducibility. Which conclusion is supported?
A) The repressor is necessary for transcription in all conditions
B) The inducer directly removes RNA polymerase from DNA
C) The reporter gene is deleted in the mutant
D) The repressor normally inhibits transcription without inducer

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Q12 An mRNA coding segment is 5′-AUG UUU GGA UAA-3′. A single base change converts UUU to UUC. With the standard code, which outcome follows?
A) The same amino acid is encoded at the second position
B) Translation stops at the second codon
C) The reading frame shifts by one nucleotide
D) A different amino acid is necessarily incorporated

Q13 During an isometric skeletal-muscle contraction, a sarcomere develops tension while its overall length is held fixed. Which molecular event still occurs?
A) Thin filaments themselves contract in length
B) A bands necessarily shorten
C) Myosin heads cycle on actin and consume ATP
D) Z discs must move closer together

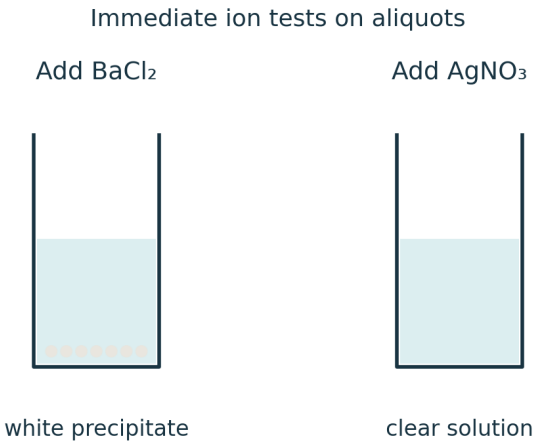
Q14 A diploid cell has 2n=8. After S phase, but before sister chromatids separate in mitosis, how many chromosomes and DNA molecules does it contain?
A) 8 chromosomes and 8 DNA molecules
B) 8 chromosomes and 16 DNA molecules
C) 16 chromosomes and 16 DNA molecules
D) 16 chromosomes and 8 DNA molecules

Q15 A young dicot stem is wounded around the vascular bundles. The surviving cells between adjacent bundles divide to form a continuous cambial ring. Which tissue provided those connecting cambial segments?
A) Sieve-tube elements of phloem
B) Mature xylem vessels
C) Cork cells outside the cortex
D) Interfascicular parenchyma

Chemistry

Q16 Both CO_2 and SO_2 contain polar bonds. Their measured molecular dipoles are zero and nonzero respectively. Which structural difference accounts for the contrast?
A) CO_2 is linear whereas SO_2 is bent
B) CO_2 has no bond polarity whereas SO_2 does
C) Both are linear but sulfur is heavier
D) SO_2 has a zero bond angle

Q17 An aqueous Co(III) coordination salt has five NH_3, one Cl and one SO_4 per cobalt. The figure shows its two immediate ion tests. Which arrangement fits those results?



A) $[Co(NH_3)_5SO_4]Cl$
B) $[Co(NH_3)_5Cl]SO_4$
C) Both Cl and SO_4 are free counterions
D) Both Cl and SO_4 are coordinated ligands

Q18 For hydrogen, the longest-wavelength line in the Balmer series is at 656 nm. What is the wavelength of the Balmer series limit, approximately, using the Rydberg relation?
A) 656 nm
B) 486 nm
C) 365 nm
D) 1312 nm

Q19 Two compounds have formula C_3H_6O. One gives a positive Tollens test, the other does not; both have a carbonyl group. Which pair can satisfy the observations?
A) Two conformers of propanone
B) Propanone and propan-2-ol
C) Propanal and propan-1-ol
D) Propanal and propanone

Q20 Complete combustion of a 0.90 g compound containing only C, H and O gives 1.32 g CO_2 and 0.54 g H_2O. What is its empirical formula? Use C=12, H=1, O=16.
A) CH_2O
B) C_2H_6O
C) CH_4O
D) C_2H_4O

Q21 An optically active secondary alkyl bromide reacts with aqueous hydroxide through a mechanism dominated by backside attack. If competing pathways are negligible, what happens at the reacting stereocentre?
A) Configuration is retained
B) Configuration inverts
C) A racemic mixture forms necessarily
D) The C–Br bond remains intact

Q22 Two colourless liquids each react with sodium metal to release H_2. Only liquid X also gives brisk CO_2 effervescence with aqueous $NaHCO_3$. Which assignment is consistent?
A) Both must be ethers
B) X is an alcohol; Y is a carboxylic acid
C) Both must be phenols
D) X is a carboxylic acid; Y can be an alcohol

Q23 A sample gives both a silver mirror with Tollens reagent and a yellow precipitate in the iodoform test. Among these compounds, which fits both results?
A) Ethanal
B) Propanal
C) Propanone
D) Benzaldehyde

Q24 In water, p-methoxyaniline is more basic than p-nitroaniline. Which electronic comparison is most directly relevant, while keeping the same solvent and aniline framework?
A) Basicity is determined only by molecular mass
B) Nitro donates by resonance more strongly than methoxy
C) Both substituents remove the nitrogen lone pair
D) Methoxy donation favours nitrogen lone-pair availability relative to nitro withdrawal

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Q25 At the same temperature, equal-molality ideal solutions of glucose and fully dissociated $CaCl_2$ are prepared in the same solvent. What is the ratio of magnitudes of their freezing-point depressions, $CaCl_2$ to glucose?
A) 1:1
B) 2:1
C) 3:1
D) 4:1

Q26 A weak-electrolyte solution is diluted tenfold. Its specific conductivity decreases, while its molar conductivity increases. Which explanation reconciles the trends?
A) Both ion concentration and dissociation per mole necessarily fall
B) Fewer ions per unit volume but greater dissociation per mole
C) The ions become chemically neutral on dilution
D) Specific and molar conductivity must always move together

Q27 The figure gives three initial-rate trials for A+B→products. From its controlled concentration changes, what is the overall reaction order?

Initial-rate measurements

Trial	[A] / M	[B] / M	Rate / units
1	0.10	0.10	2
2	0.20	0.10	8
3	0.20	0.20	16

A) 1
B) 2
C) 3
D) 4

Q28 For a reaction, pathway A has $\Delta H_1 = -80$ kJ mol^{-1} followed by $\Delta H_2 = +30$ kJ mol^{-1}. A direct pathway B reaches the same final state. What is ΔH for B, and why?
A) −110 kJ mol^{-1} because signs add in magnitude
B) −50 kJ mol^{-1} because enthalpy is a state function
C) +50 kJ mol^{-1} because the second step is endothermic
D) The value depends on activation energy

Q29 A buffer contains 0.10 mol CH_3COOH and 0.10 mol CH_3COO^- in 1 L. After 0.02 mol strong acid is added with negligible volume change, what is the new pH? Use $pK_a = 4.76$ and $\log(2/3) = -0.176$.
A) 4.58
B) 4.76
C) 4.94
D) 5.11

Q30 Two high-spin octahedral ions have respectively three and five unpaired electrons. In the spin-only approximation, what is the ratio of their magnetic moments, first to second?
A) $3 : 5$
B) $\sqrt{5} : \sqrt{7}$
C) $\sqrt{3} : \sqrt{5}$
D) $\sqrt{15} : \sqrt{35}$

Mathematics

Q31 A circle $x^2 + y^2 - 6x + 2y - 6 = 0$ is cut by the line $y = 1$. If the line is shifted upward by 2 units, by how much does the length of its chord decrease?
A) 8
B) $2\sqrt{15}$
C) $2\sqrt{3}$
D) $4\sqrt{3}$

Q32 Let $f(x) = x^2 \sin(1/x)$ for $x \neq 0$, and $f(0) = 0$. For which real k is $g(x) = f(x) + k|x|$ differentiable at zero?
A) Every k
B) Only k=1
C) Only k=0
D) No k

Q33 Find the maximum of $f(x) = x^3 - 3x$ on the closed interval $[-2, 1]$.
A) 2
B) 0
C) −2
D) 3

Q34 Evaluate $I = \int_{-1}^{1} \left(\frac{x^3+2x}{1+x^2} + \frac{2}{1+x^2} \right) dx$.
A) 0
B) π
C) π/2
D) 2π

Q35 On the domain [1,∞), let f(x)=x+1/x. Which formula gives its inverse, including the correct domain of the inverse?
A) $(y + \sqrt{y^2 - 4})/2$, y≥2
B) $(y - \sqrt{y^2 - 4})/2$, y≥2
C) $(y + \sqrt{y^2 + 4})/2$, y≥2
D) $y - 1/y$, y≥2

Q36 Vectors **a** $= (1, 2, 0)$, **b** $= (0, 1, 1)$ and **c** $= (t, 0, 1)$ are coplanar for what value of t?
A) −2
B) 2
C) 1
D) −1/2

Q37 For which real a is $A = \begin{pmatrix} 1 & a & 0 \\ 0 & 1 & a \\ a & 0 & 1 \end{pmatrix}$ singular, and what spans its nullspace at that value?
A) a=−1; (1,1,1)
B) a=−1; (1,−1,1)
C) a=1; (1,1,1)
D) It is nonsingular for all real a

Q38 A box contains 4 red and 3 blue balls. Two are drawn without replacement. Given that at least one is blue, what is the probability both are blue?
A) 1/7
B) 1/3
C) 1/5
D) 3/7

Q39 Six distinct people sit round a table. Two specified people may not sit adjacent; rotations are identical and reflections distinct. How many arrangements exist?
A) 48
B) 72
C) 96
D) 120

Q40 What is the coefficient of x^4 in $(1-x)^5(1+x)^4$?
A) -6
B) -1
C) 1
D) 6

Q41 A decreasing geometric progression has positive first term. The sum of its first three terms is 7, and the sum of its next three terms is 7/8. What is the sum to infinity?
A) 8
B) 7
C) 14
D) 16

Q42 The quadratics x^2-3x+2 and $x^2-(a+1)x+a^2-1$ share at least one real root. What is the sum of all distinct real values of a that allow this?
A) 1
B) 3
C) 2
D) $(1+\sqrt{5})/2$

Q43 A dataset has mean 5 and standard deviation 3. Each observation becomes $y=2x+c$, and the new mean is 17. What is the new standard deviation?
A) 3
B) 9
C) 6
D) 12

Q44 A positive solution satisfies $dy/dx=xy^2$, y(0)=1. At which positive x does it first reach y=2?
A) 1/2
B) 2
C) √2
D) 1

Q45 For $f(x)=|x-1|/(x^2-1)$, x≠1, what are its left and right limits at x=1, in order?
A) (1/2,1/2)
B) (−1/2,1/2)
C) (1/2,−1/2)
D) (−1/2,−1/2)

Physics

Q46 Measured length and width are (2.40±0.02) cm and (1.50±0.01) cm. Find the maximum first-order percentage uncertainty in their product.
A) 0.67%
B) 0.83%
C) 1.50%
D) 2.00%

Q47 Masses 2 kg and 3 kg hang over a light pulley whose support accelerates upward at 2 m s^{-2}. Relative to the support, find their acceleration magnitude. Take g=10 m s^{-2}.
A) 1
B) 2
C) 2.4
D) 4

Q48 A wire within its elastic limit has length L, area A and strain ε. If L and A are both doubled while ε is kept unchanged, by what factor does stored elastic energy change?
A) 1
B) 2
C) 8
D) 4

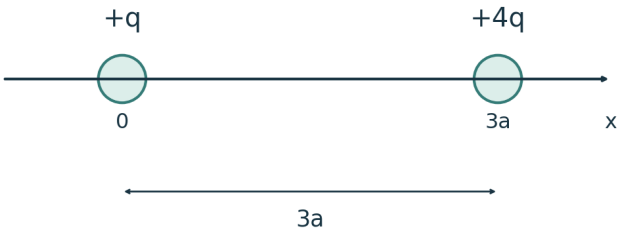
Q49 Two slabs of the same cross-section are in steady series heat flow. Slab 1 has thickness 2d and conductivity 2k; slab 2 thickness d and conductivity k. What fraction of the total temperature drop is across slab 1?
A) 1/4
B) 1/2
C) 2/3
D) 3/4

Q50 At fixed temperature, a dilute ideal gas's pressure doubles while molecular collision diameter is unchanged. How do mean free path and mean molecular speed change?
A) Both double
B) Both halve
C) Path unchanged; speed doubles
D) Path halves; speed unchanged

Q51 A particle has position-dependent acceleration $a = 2x$ in SI units. At x=1 m its speed in +x is 3 m s^{-1}. Find its speed at x=2 m.
A) √11
B) √15
C) 5
D) 7

Q52 A string pulse reflects inverted from a junction with a much heavier string. What boundary behaviour explains the inversion?
A) Approximately fixed end
B) Completely free end
C) Negative wave frequency
D) Faster reflected wave

Q53 The figure marks +q at x=0 and +4q at x=3a. Where between them is the net electric field zero?



A) Nowhere
B) 3a/2
C) 2a
D) a

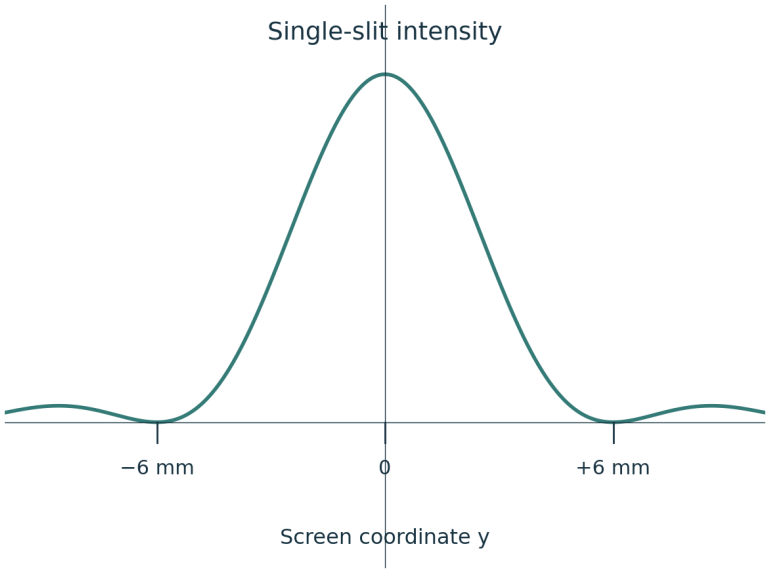
Q54 A balanced Wheatstone bridge has P=100 Ω, Q=200 Ω, R=50 Ω, S=100 Ω. R warms to 60 Ω; all other arms initially remain fixed. To restore balance, what must S become?
A) 60 Ω
B) 100 Ω
C) 120 Ω
D) 240 Ω

Q55 A conducting rod rotates about one end at angular speed ω in perpendicular uniform field B. If only its outer half, r∈[L/2,L], is conducting, what fraction of full-rod emf is across that segment?
A) 1/4
B) 1/2
C) 3/4
D) 1

Q56 A series AC circuit has R=30 Ω, XL=40 Ω and XC=30 Ω. With 100 V rms supply, what is rms current?
A) 2 A
B) √10 A
C) 10/3 A
D) 5 A

Q57 A converging lens produces an upright virtual image of an object inside its focal length. In the Cartesian convention with incident light left-to-right, what signs have image distance v and magnification m?
A) v=0 always
B) v>0, m>0
C) v>0, m<0
D) v<0, m>0

Q58 The figure places first minima of a single-slit pattern at y=±6 mm. If the slit width doubles with wavelength and screen distance unchanged, what is the new full central maximum width?



A) 3 mm
B) 6 mm
C) 12 mm
D) 24 mm

Q59 A metal emits at 400 nm with stopping potential 1.0 V but not at 600 nm. With hc=1240 eV·nm, find work function and threshold wavelength.
A) 2.10 eV, ≈590 nm
B) 3.10 eV, 400 nm
C) 1.0 eV, 1240 nm
D) 4.10 eV, ≈300 nm

Q60 Two identical p-n diodes have equal series resistors on the same DC supply. X has p-side at higher potential; Y has n-side at higher potential. Before breakdown, which comparison holds?
A) X conducts appreciably; Y has small reverse current
B) Both conduct equally
C) Y conducts appreciably; X is reverse biased
D) Neither can conduct

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / A taxonomic hierarchy

Answer: (A) The first two share a genus, and all three share a family.

Why: A species is nested within a genus, and genera are nested within a family. Different species do not share a species designation; the third organism's different genus leaves family as the most specific rank common to all three. This uses the stated ranks, without assuming that close resemblance changes formal placement.

Why not the others: (B) downgrades the first pair despite their shared genus; (C) incorrectly merges different species and genera; (D) places the third in a genus explicitly excluded.

Key point: Compare the lowest common rank in a nested hierarchy.

Q2 / C root vascular arrangement

Answer: (C) The first is a dicot root and the second a monocot root.

Why: Radial vascular tissue with xylem arms alternating with phloem establishes root anatomy. A small or absent central pith and a few xylem arms typify the dicot root; numerous alternating strands around a developed pith support the monocot root. The centre and arrangement must be read together, not a single coloured region.

Why not the others: (A) treats the large pith and many vascular units as an age effect; (B) misreads the first radial stele as a stem; (D) misreads the first section as a dicot stem despite its radial root pattern.

Key point: Use both vascular arrangement and pith.

Q3 / B ribosome translation context

Answer: (B) 27 fewer amino acids.

Given: 75 sense codons originally and 48 before the premature stop.

Formula/Principle: Each sense codon adds one amino acid; a stop codon adds none.

Step 1: Original length is 75 amino acids and mutant length is 48.

Step 2: Subtract $75-48=27$.

Why: The ribosome reads successive triplets on the mRNA; termination releases the chain at the new stop without attaching an amino acid for that stop. Counting the start codon among the stated sense codons is essential.

Why not the others: (A) wrongly counts one of the stop positions; (C) adds an extra residue; (D) reports the original length instead of the loss.

Key point: Count sense codons, not the termination triplet.

Q4 / A chemiosmotic gradient

Answer: (A) The carrier dissipates the transmembrane proton gradient.

Why: Electron flow can still oxidise water and release oxygen, while the carrier allows protons to return without passing through ATP synthase. The resulting loss of proton motive force deprives ATP synthase of its driving gradient. A short period of oxygen evolution is consistent with an electron path that has not itself been blocked.

Why not the others: (B) would stop oxygen formation directly; (C) is not a membrane carrier's action; (D) reverses the energy conversion of ATP synthase.

Key point: Separate electron transport from coupling through the proton gradient.

Q5 / B endocrine feedback

Answer: (B) Both TRH and TSH tend to rise.

Why: Thyroxine normally inhibits hypothalamic TRH and pituitary TSH through negative feedback. A fall in thyroid hormone removes part of this inhibition at both upstream levels, so the intact axis increases drive to the thyroid. The premise rules out a primary pituitary failure, which would change the inference.

Why not the others: (A) applies inhibition to the hypothalamus despite low hormone; (C) inhibits TSH in the wrong direction; (D) is the usual response to excess rather than deficient thyroid hormone.

Key point: Follow the direction of negative feedback.

Q6 / C pollen compatibility and outbreeding

Answer: (C) Self-incompatibility blocks tube growth on the self stigma.

Why: Hydration confirms that pollen arrived and was initially viable on both stigmas. The first observed divergence is tube growth within the style; a genetically controlled pollen-pistil interaction can reject self pollen at or after germination while accepting compatible pollen of the same species. The diagram distinguishes arrival from successful fertilisation.

Why not the others: (B) contradicts deposition on both stigmas; (A) contradicts hydration and the successful cross; (D) occurs far later and cannot explain absent pollen tubes.

Key point: Locate the earliest incompatible step in the pollen-pistil pathway.

Q7 / D molecular replication direction

Answer: (D) A new 5′ to 3′ strand extends rightward.

Why: DNA polymerase reads the top template in the 3′ to 5′ direction as it travels right, adding nucleotides to the 3′ end of the complementary strand. Its 5′ to 3′ synthesis therefore keeps pace with the rightward fork. Direction is fixed by template polarity, not by which strand is drawn above the other.

Why not the others: (B) violates polymerase synthesis polarity; (C) describes synthesis away from this fork; (A) ignores complementarity and the available template.

Key point: Read template 3′ to 5′ to determine leading-strand direction.

Q8 / A antibiotic resistance selection

Answer: (A) The allele was present before treatment.

Why: Detection of the identical resistance allele in an archived pretreatment sample establishes that the variant preceded the selection pressure. Antibiotic exposure can enrich this pre-existing variant because susceptible cells die or divide less. Growth differences after treatment do not establish when the mutation first arose.

Why not the others: (B) concerns fitness cost, not mutation timing; (C) describes drug action but not origin of the allele; (D) records population size without revealing ancestry.

Key point: A pre-exposure control separates selection from induced origin.

Q9 / C restriction recognition and ends

Answer: (C) Blunt fragments can ligate without complementary overhang pairing.

Why: DNA ligase seals a phosphodiester bond when ends are juxtaposed; cohesive overhangs make this pairing more favourable and can influence compatibility. A blunt cut removes the overhang sequence that formerly guided annealing, though it does not abolish ligation. Directional insertion would require distinct compatible ends, which the changed setup does not specify.

Why not the others: (B) treats inefficient blunt ligation as impossible; (A) assumes orientation control absent here; (D) contradicts the stated change of enzyme.

Key point: Distinguish end compatibility from ligase chemistry.

Q10 / B population growth regime

Answer: (B) The per-capita rate is higher at half the carrying capacity.

Why: The logistic model gives $\frac{1}{N}\frac{dN}{dt} = r(1 - N/K)$. At $K/2$ the factor is $1/2$, whereas at K it is zero because density limitation balances growth. This asks for growth per individual, not the total change in population size.

Why not the others: (A) ignores zero net growth at capacity; (C) assumes density has no effect; (D) incorrectly transfers the capacity boundary to half capacity.

Key point: Divide logistic growth by N before comparing per-capita rates.

Q11 / D transcription regulation

Answer: (D) Repressor action keeps the operon off without inducer.

Why: The deletion is a loss-of-function comparison: expression without inducer appears when repressor is absent and the inducible pattern returns when it is restored. That rescue links the phenotype to repressor-mediated negative control, rather than an unrelated mutation. It does not by itself identify every molecular contact of the inducer.

Why not the others: (B) predicts decreased expression on induction; (C) conflicts with the reporter signal; (A) conflicts with expression in the repressor-null strain.

Key point: Use a knockout and restoration together to infer regulatory direction.

Q12 / A genetic-code reading

Answer: (A) Both UUU and UUC encode phenylalanine.

Why: Reading begins at AUG and proceeds in non-overlapping triplets. The substitution changes the third base of the second codon, but these two triplets are synonymous in the standard genetic code; subsequent GGA and the UAA stop stay in frame. A base change need not change protein sequence.

Why not the others: (B) confuses UUC with a stop codon; (C) requires an insertion or deletion, not a substitution; (D) ignores codon degeneracy.

Key point: Translate the altered codon before predicting protein effects.

Q13 / C skeletal muscle contraction

Answer: (C) Cross-bridge cycling can generate force without net shortening.

Why: Isometric means the externally measured sarcomere length is held fixed, not that contractile proteins become inactive. Actin-myosin attachment, power strokes and ATP-dependent detachment can continue against an opposing load while net displacement remains zero. Thick and thin filament lengths do not change during sliding-filament action.

Why not the others: (B) changes thick-filament length; (A) changes actin length; (D) would indicate shortening rather than the stated fixed length.

Key point: Separate force generation from net movement.

Q14 / B chromosome versus DNA content

Answer: (B) Eight chromosomes and sixteen DNA molecules.

Given: Diploid chromosome count is 8; DNA has replicated and centromeres have not separated.

Formula/Principle: Chromosomes are counted by centromeres; each chromatid contains one DNA molecule.

Step 1: Replication doubles chromatids per chromosome from one to two without doubling centromeres.

Step 2: Count 8 centromeres and $8 \times 2 = 16$ chromatids/DNA molecules.

Why: S phase copies DNA, while sister chromatids stay joined at their centromeres until anaphase. A chromosome can therefore consist of two DNA molecules at this stage. The stated timing is what distinguishes DNA content from chromosome number.

Why not the others: (A) ignores S phase; (C) counts chromatids as separate chromosomes too early; (D) reverses the two counts.

Key point: Count centromeres and DNA duplexes separately.

Q15 / D stem cambium origin

Answer: (D) Interfascicular parenchyma contributes the connecting cambium.

Why: Within a bundle the original cambium is intrafascicular. Between bundles, living parenchyma cells regain division capacity and form interfascicular cambium, joining the segments into a continuous ring for secondary growth. The question specifically concerns the bridge between vascular bundles rather than the pre-existing cambium inside one bundle.

Why not the others: (B) has dead vessel elements unsuitable for division; (C) lies outside the relevant ring; (A) is a specialised conducting element, not the intervening parenchyma.

Key point: Identify where secondary meristem arises between bundles.

Chemistry

Q16 / A bond polarity versus molecular dipole

Answer: (A) CO_2 is linear and SO_2 is bent.

Why: Bond-dipole vectors in linear CO_2 are equal and oppositely directed, so their sum vanishes. The bent arrangement of SO_2 prevents the two S–O bond dipoles from cancelling completely. A molecular dipole is a vector sum, so polar bonds alone do not determine whether the molecule is polar.

Why not the others: (B) contradicts polar C–O bonds; (C) neglects geometry; (D) confuses bent geometry with a zero angle.

Key point: Add bond dipoles as vectors using molecular shape.

Q17 / B ionisation isomer wet test

Answer: (B) $[Co(NH_3)_5Cl]SO_4$.

Why: Free sulfate precipitates as $BaSO_4$, while coordinated chloride does not supply immediate free Cl^- for AgCl. In (B), Co(III) plus one bound Cl^- gives the complex charge +2, balanced by one SO_4^{2-}. The wet tests and electroneutrality agree; moving sulfate inside and chloride outside reverses the ionic responses.

Why not the others: (A) predicts AgCl instead of $BaSO_4$; (C) predicts both precipitates; (D) predicts neither.

Key point: Check free ions and complex charge together.

Q18 / C spectral series limit

Answer: (C) About 365 nm.

Given: The 3→2 line has 656 nm; the series limit is ∞→2.

Formula/Principle: $1/\lambda = R(1/2^2 - 1/n^2)$.

Step 1: For 3→2, the factor is $1/4 - 1/9 = 5/36$; at the limit it is $1/4$.

Step 2: $\lambda_{\infty\to 2} = 656(5/36)/(1/4) = 656(5/9) \approx 364.4$ nm.

Worked derivation:

$$\lambda_{\text{limit}} = 656\frac{1/4 - 1/9}{1/4} = 656\frac{5}{9} \approx 364.4 \text{ nm}.$$

Why: A larger transition energy gives a shorter wavelength. Taking $n \to \infty$ yields the maximum energy of lines ending at level two, so the limit must lie below the 656 nm first line.

Why not the others: (B) is another finite Balmer transition; (A) keeps the first line; (D) doubles its wavelength.

Key point: Compare reciprocal wavelengths through their level factors.

Q19 / D isomer classification

Answer: (D) Propanal and propanone are functional isomers.

Why: Both have the formula and a carbonyl group, but the aldehyde propanal is normally oxidised by Tollens reagent while the ketone propanone is not. Changing the carbonyl position from terminal aldehyde to internal ketone changes the functional group and the wet-test outcome without changing molecular formula.

Why not the others: (A) supplies the same ketone twice, so Tollens outcomes would agree; (B) includes an alcohol without the stipulated carbonyl; (C) also includes an alcohol and cannot explain two carbonyl compounds.

Key point: Combine molecular formula with a functional-group reaction.

Q20 / A combustion composition

Answer: (A) CH_2O.

Given: $m(CO_2) = 1.32$ g, $m(H_2O) = 0.54$ g, sample 0.90 g.

Formula/Principle: Carbon and hydrogen moles come from combustion products; oxygen follows by mass difference.

Step 1: $n_C = 1.32/44 = 0.030$, $n_H = 2(0.54/18) = 0.060$ mol.

Step 2: Oxygen mass $0.90 - 0.36 - 0.06 = 0.48$ g, hence $n_O = 0.48/16 = 0.030$ mol.

Step 3: Divide by 0.030 to obtain C:H:O = 1:2:1.

Worked derivation:

$$n_C = 1.32/44 = 0.030, \quad n_H = 2(0.54/18) = 0.060,$$
$$n_O = (0.90 - 0.36 - 0.060)/16 = 0.030 \text{ mol},$$
$$C : H : O = 1 : 2 : 1 \Rightarrow CH_2O.$$

Why: Combustion converts all sample carbon to carbon dioxide and hydrogen to water under the stated complete-combustion assumption. Oxygen cannot be inferred from the CO_2 mass directly because air supplies oxygen too; mass balance isolates sample oxygen.

Why not the others: (B) is not the smallest ratio; (C) doubles hydrogen; (D) gives too much carbon.

Key point: Use product moles for C/H and mass balance for O.

Q21 / B stereochemical substitution

Answer: (B) Backside substitution inverts configuration.

Why: In an SN2 event the nucleophile approaches opposite the departing bromide, and bond making and breaking occur in one step. The tetrahedral arrangement turns inside out at that carbon. Whether the R/S letter changes depends also on priority order of substituents; the geometrical statement of inversion is the reliable inference here.

Why not the others: (A) would require a different pathway or special sequence; (C) is associated with a planar intermediate and possible attack from both sides; (D) contradicts substitution.

Key point: Distinguish stereochemical inversion from an R/S label change.

Q22 / D sodium versus neutral reagent response

Answer: (D) X is a carboxylic acid; Y may be an alcohol.

Why: Sodium reacts with acidic O–H groups in both alcohols and carboxylic acids, so its positive test alone is not diagnostic. Bicarbonate is protonated by a carboxylic acid to yield carbonic acid and then CO_2; ordinary alcohols are too weakly acidic to give that reaction. The second reagent separates the possibilities.

Why not the others: (B) reverses the bicarbonate response; (C) phenols ordinarily do not liberate CO_2 from bicarbonate; (A) ethers lack the O–H group needed for the sodium test.

Key point: Use a discriminating second reagent after a non-specific first test.

Q23 / A Tollens versus iodoform evidence

Answer: (A) Ethanal.

Why: Tollens reagent is reduced by the aldehyde group. The iodoform reaction requires a methyl carbonyl unit or a precursor oxidised to one; ethanal, CH_3CHO, is the relevant aldehyde with this motif. Comparing the two tests excludes a methyl ketone that passes only the second and other aldehydes that lack the required methyl group.

Why not the others: (B) propanal lacks a methyl directly attached to the carbonyl; (C) propanone is a methyl ketone but normally fails Tollens; (D) benzaldehyde lacks the methyl-carbonyl unit.

Key point: Intersect the structural conditions of two wet tests.

Q24 / D basicity in solvent

Answer: (D) Para methoxy donation favours basicity over nitro withdrawal.

Why: The aniline nitrogen lone pair participates in conjugation with the ring. An electron-donating para substituent and a strongly withdrawing para nitro group shift the electron distribution in opposite directions; the latter reduces availability for protonation. Holding solvent and core structure fixed makes the comparison meaningful, although solvation also matters in absolute basicity.

Why not the others: (B) reverses substituent effects; (C) falsely removes a lone pair; (A) ignores proton affinity and solvation.

Key point: Compare substituent effects under the same solvent conditions.

Q25 / C colligative-property comparison

Answer: (C) 3:1.

Given: Equal molality and the same solvent; glucose is non-electrolytic, $CaCl_2$ fully dissociates.

Formula/Principle: $\Delta T_f = iK_f m$.

Step 1: Glucose has $i = 1$; $CaCl_2 \longrightarrow Ca^{2+} + 2\,Cl^-$ gives three ideal particles, so $i = 3$.

Step 2: $\Delta T_{f,CaCl_2}/\Delta T_{f,glucose} = 3K_f m/(K_f m) = 3$.

Worked derivation:

$$CaCl_2 \longrightarrow Ca^{2+} + 2\,Cl^-, \quad \frac{\Delta T_{f,\text{salt}}}{\Delta T_{f,\text{glucose}}} = \frac{3K_f m}{K_f m} = 3.$$

Why: Freezing-point depression depends on the number of dissolved particles, not their chemical identity in the ideal limit. Equal solvent constants and molality cancel; complete dissociation supplies the remaining factor. Real ionic solutions may deviate, but complete ideal dissociation is stipulated.

Why not the others: (A) ignores dissociation; (B) counts only the two chloride ions; (D) adds an extra particle.

Key point: Count all solute ions in the ideal van 't Hoff factor.

Q26 / B conductivity dilution

Answer: (B) Ion density falls while degree of dissociation rises.

Why: Specific conductivity concerns conductance of a fixed geometry/volume, so dilution usually reduces charge carriers per unit volume. Molar conductivity scales conductance to one mole of electrolyte; increased dissociation of the weak electrolyte supplies more ions per initial formula unit, raising this quantity. The two measurements use different normalisations.

Why not the others: (A) predicts the wrong molar trend; (C) has no basis in simple dilution; (D) conflates volume and mole normalisation.

Key point: State what each conductivity is normalised to.

Q27 / C initial-rate order table

Answer: (C) Third order.

Given: Doubling A at fixed B multiplies rate by four; doubling B at fixed A multiplies rate by two.

Formula/Principle: $v = k[A]^m[B]^n$.

Step 1: $2^m = 8/2 = 4$, so $m = 2$.

Step 2: $2^n = 16/8 = 2$, so $n = 1$, and $m + n = 3$.

Worked derivation:

$$2^m = 8/2 = 4 \Rightarrow m = 2, \quad 2^n = 16/8 = 2 \Rightarrow n = 1, \quad m + n = 3.$$

Why: Each rate ratio is taken under a controlled change to just one reactant, isolating its exponent. Reading the first and third trials directly would change both concentrations and confound the two orders. The illustrated table tests the power law experimentally rather than using stoichiometric coefficients.

Why not the others: (A) misses A's square dependence; (B) omits one exponent; (D) adds an unsupported power.

Key point: Vary one concentration at a time.

Q28 / B Hess-cycle route

Answer: (B) $-50\,\mathrm{kJ\,mol^{-1}}$.

Given: Two consecutive steps have enthalpy changes −80 and +30 kJ mol⁻¹.

Formula/Principle: Hess's law: total state change equals the sum along any path.

Step 1: $\Delta H_A = (-80) + (+30) = -50\,\mathrm{kJ\,mol^{-1}}$.

Step 2: Direct path B has the same initial/final states, so $\Delta H_B = -50\,\mathrm{kJ\,mol^{-1}}$.

Worked derivation:

$$\Delta H_B = \Delta H_A = (-80) + 30 = -50\ \mathrm{kJ\,mol^{-1}}.$$

Why: Enthalpy depends on endpoints, not the reaction route. The second step partially offsets the first exothermic change; it does not reverse the net sign. A catalyst could alter activation energies while leaving this endpoint difference intact.

Why not the others: (A) mishandles signed addition; (C) ignores the larger first step; (D) confuses kinetics with thermodynamics.

Key point: Add signed enthalpy changes along a route.

Q29 / A buffer before and after neutralisation

Answer: (A) Approximately 4.58.

Given: Initial acid/base each 0.10 mol; 0.02 mol H^+ reacts with acetate.

Formula/Principle: pH $= pK_a + \log(n_{base}/n_{acid})$ after stoichiometric neutralisation.

Step 1: $n_{base} = 0.10 - 0.02 = 0.08$; $n_{acid} = 0.10 + 0.02 = 0.12$ mol.

Step 2: Ratio $0.08/0.12 = 2/3$; pH $= 4.76 - 0.176 = 4.584$.

Worked derivation:

$$\begin{aligned} n_{\text{base}} &= 0.10 - 0.02 = 0.08, \quad n_{\text{acid}} = 0.10 + 0.02 = 0.12, \\ pH &= 4.76 + \log_{10}(0.08/0.12) \\ &= 4.76 - 0.176 = 4.584. \end{aligned}$$

Why: Strong acid first converts acetate to undissociated acetic acid; the Henderson ratio must use the post-reaction amounts. Negligible volume change lets mole ratio replace concentration ratio. Buffering limits the pH shift but does not leave pH fixed.

Why not the others: (B) uses the initial 1:1 buffer; (C) reverses the ratio; (D) overestimates the pH.

Key point: Complete strong-acid neutralisation before applying the buffer relation.

Q30 / D spin-only moment

Answer: (D) $\sqrt{15} : \sqrt{35}$, equivalently $\sqrt{3} : \sqrt{7}$.

Given: Unpaired counts are $n_1 = 3$ and $n_2 = 5$.

Formula/Principle: Spin-only $\mu = \sqrt{n(n+2)}$ Bohr magnetons.

Step 1: $\mu_1 = \sqrt{3(5)} = \sqrt{15}$; $\mu_2 = \sqrt{5(7)} = \sqrt{35}$.

Step 2: Divide, $\mu_1/\mu_2 = \sqrt{15/35} = \sqrt{3/7}$.

Worked derivation:

$$\mu_1 = \sqrt{3(5)}\,\mu_B, \quad \mu_2 = \sqrt{5(7)}\,\mu_B, \quad \frac{\mu_1}{\mu_2} = \sqrt{\frac{3}{7}}.$$

Why: The common Bohr-magneton unit cancels in the ratio; counting unpaired electrons alone is insufficient because the spin-only relation contains $n+2$. The high-spin condition ensures the stated unpaired counts rather than a low-spin configuration.

Why not the others: (B) keeps only the n+2 factors; (C) drops the +2 terms; (A) assumes a linear dependence.

Key point: Use $n(n+2)$, then simplify the ratio.

Mathematics

Q31 / D chord distance

Answer: (D) $4\sqrt{3}$.

Given: The original and shifted lines are y=1 and y=3.

Formula/Principle: For a circle of radius r, a line at centre distance d cuts a chord of length $2\sqrt{r^2 - d^2}$.

Step 1: Completing squares gives $(x-3)^2 + (y+1)^2 = 16$, centre (3,−1) and radius 4.

Step 2: Distances to y=1 and y=3 are 2 and 4, so the latter is tangent and its chord has length zero.

Worked derivation:

$$(x-3)^2+(y+1)^2=16,\quad L_1=2\sqrt{16-2^2}=4\sqrt{3},\quad L_3=2\sqrt{16-4^2}=0,\quad L_1-L_3=4\sqrt{3}.$$

Why: The shifted line reaches the top tangent at y=3. Checking that boundary prevents treating it as a second ordinary secant. The requested decrease is the original chord minus the new zero-length contact.

Why not the others: (A) treats the line as passing through the centre; (B) uses centre distance one instead of two; (C) computes only half the actual original chord.

Key point: Complete squares, then check whether a shifted line is secant or tangent.

Q32 / C parameterised differentiability

Answer: (C) Only $k=0$.

Given: Oscillatory term is multiplied by x^2; the modulus creates opposite one-sided slopes.

Formula/Principle: $g'(0)=\lim_{h\to0}[g(h)-g(0)]/h$.

Step 1: Quotient for f is $h\sin(1/h)\to0$ by squeeze.

Step 2: Quotient for k|h| is k for h>0 and −k for h<0; equality requires k=0.

Worked derivation:

$$\frac{g(h)-g(0)}{h}=h\sin(1/h)+k\frac{|h|}{h};\quad g'_+(0)=k,\quad g'_-(0)=-k\Rightarrow k=0.$$

Why: The oscillation in f itself has no fixed limit, but its difference quotient has an extra factor h and vanishes. The cusp term survives, so continuity of g alone is insufficient to prove differentiability.

Why not the others: (A) overlooks the cusp; (B) leaves slopes +1 and −1; (D) overlooks the valid k=0 case.

Key point: Compare one-sided difference quotients after squeezing oscillatory factors.

Q33 / A extreme value with endpoints

Answer: (A) 2.

Given: Closed interval includes both endpoints.

Formula/Principle: A continuous function reaches extrema at critical points or endpoints.

Step 1: $f'(x)=3x^2-3=0$ at x=−1,1; x=1 is also an endpoint.

Step 2: $f(-2)=-2$, $f(-1)=2$, $f(1)=-2$; choose 2.

Worked derivation:

$$f'(x)=3(x^2-1),\quad \{f(-2),f(-1),f(1)\}=\{-2,2,-2\},\quad \max_{[-2,1]}f=2.$$

Why: A stationary-point-only shortcut happens to find the maximum here but does not prove it on a closed interval; the endpoints must also be evaluated. The two endpoint values match but neither exceeds the interior value.

Why not the others: (B) is not attained as the maximum; (C) is the minimum; (D) comes from an incorrect stationary value.

Key point: Test endpoints even when a critical point is present.

Q34 / B symmetry of definite integral

Answer: (B) π.

Given: The interval is symmetric about zero.

Formula/Principle: The first rational term is odd; its integral cancels, while the even term integrates by arctangent.

Step 1: $(x^3+2x)/(1+x^2)$ is odd, so its integral on [−1,1] is zero.

Step 2: $\int_{-1}^{1} 2/(1+x^2)dx = 2[\arctan x]_{-1}^{1} = \pi$.

Worked derivation:

$$I = 0 + 2(\arctan 1 - \arctan(-1)) = 2(\pi/4 + \pi/4) = \pi.$$

Why: Reversing x changes the first term's sign but leaves the second unchanged. This odd/even decomposition avoids an unnecessary antiderivative for the odd part, then uses both bounds for the even part.

Why not the others: (A) cancels the even contribution too; (C) misses a factor two; (D) doubles the even result.

Key point: Decompose integrands by parity on symmetric intervals.

Q35 / A inverse restriction

Answer: (A) $(y+\sqrt{y^2-4})/2$ on y≥2.

Given: x≥1; f'(x)=1−1/x²≥0, with equality only at the endpoint.

Formula/Principle: The inverse exists because f increases from f(1)=2 to infinity; solve y=x+1/x and select the allowed branch.

Step 1: Multiply by x: x²−yx+1=0.

Step 2: Quadratic formula gives x=(y±√(y²−4))/2.

Step 3: The minus branch is ≤1 for y>2; select plus for x≥1.

Worked derivation:

$$f([1,\infty)) = [2,\infty), \quad x^2 - yx + 1 = 0 \Rightarrow x_{\pm} = \frac{y \pm \sqrt{y^2-4}}{2}, \quad f^{-1}(y) = x_{+}.$$

Why: The algebra produces two reciprocal roots, but only one lies in the stated original domain except at y=2 where they coincide. The domain restriction is therefore part of the inverse, not a decorative detail; monotonicity proves uniqueness.

Why not the others: (B) chooses the root below one; (C) changes the discriminant sign; (D) confuses rearranging a reciprocal with solving the quadratic.

Key point: Determine range and select the inverse branch using the original domain.

Q36 / D scalar triple product

Answer: (D) $t = -1/2$.

Given: Three vectors in 3D are coplanar iff their scalar triple product is zero.

Formula/Principle: $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$.

Step 1: $\mathbf{b} \times \mathbf{c} = (1, t, -t)$.

Step 2: Dot with a gives $1+2t$, so t=−1/2.

Worked derivation:

$$\det \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ t & 0 & 1 \end{pmatrix} = 1 + 2t = 0 \Rightarrow t = -\tfrac{1}{2}.$$

Why: Vanishing oriented volume is the exact coplanarity condition. No numerical length or angle conditions need to be introduced. The determinant vanishes precisely when the parallelepiped volume collapses, so this criterion tests a geometric property rather than a numerical coincidence.

Why not the others: (A) gives determinant −3; (B) gives determinant 5; (C) gives determinant 3. None is zero.

Key point: Compute the determinant before selecting an option.

Q37 / A inverse existence

Answer: (A) a=−1, nullspace span(1,1,1).

Given: Singularity means determinant zero; nullspace solves Av=0 at that parameter.

Formula/Principle: Expand determinant and factor over the reals.

Step 1: $\det A = 1 + a^3 = (a+1)[(a-1/2)^2 + 3/4]$, so a=−1 only.

Step 2: At a=−1, row equations are x−y=0, y−z=0 and −x+z=0.

Step 3: Thus x=y=z, giving v=t(1,1,1), t real.

Worked derivation:

$$\det A = 1 + a^3 = 0 \iff a = -1; \qquad A(-1)(x,y,z)^T = 0 \iff x = y = z.$$

Why: A zero determinant locates the singular parameter but does not determine the nullspace direction. Solving the dependent equations supplies that independent piece; the positive quadratic factor rules out extra real parameters.

Why not the others: (B) its vector fails x=y; (C) a=1 gives determinant 2; (D) overlooks the real singular value.

Key point: Follow the determinant test with a nullspace solve.

Q38 / C without-replacement conditional event

Answer: (C) 1/5.

Given: The condition excludes the two-red outcomes.

Formula/Principle: Conditional probability is $P(BB)/[1 - P(RR)]$.

Step 1: $P(BB) = (3/7)(2/6) = 1/7$, $P(RR) = (4/7)(3/6) = 2/7$.

Step 2: Divide $(1/7)/(5/7) = 1/5$.

Worked derivation:

$$P(BB \mid B \geq 1) = \frac{(3/7)(2/6)}{1 - (4/7)(3/6)} = \frac{1}{5}.$$

Why: The draws are dependent, and 'at least one' restricts the sample space rather than fixing the first draw. Checking the unconditional both-blue probability against the conditional result confirms that restricting to at-least-one-blue increases the relative share of blue-blue outcomes.

Why not the others: (A) is unconditional; (B) assumes blue first; (D) ignores the conditional denominator.

Key point: Rebuild the sample space after conditioning.

Q39 / B circular arrangement with forbidden pair

Answer: (B) 72.

Given: Six labelled seats up to rotation.

Formula/Principle: All circular arrangements minus adjacent-pair block arrangements.

Step 1: Total 5!=120.

Step 2: Forbidden 2·4!=48; allowed 72.

Worked derivation:

$$N = 5! - 2(4!) = 120 - 48 = 72.$$

Why: Treating the pair as a block includes adjacency across a cut in a linear listing. Two internal pair orders are distinct; mirror images are also distinct by the stated rule.

Why not the others: (A) counts forbidden cases; (C) subtracts one block order; (D) ignores the restriction.

Key point: Count circular adjacency by a block and subtract.

Q40 / D signed coefficient extraction

Answer: (D) 6.

Given: Four pairs of conjugate factors are present.

Formula/Principle: $(1-x)^5(1+x)^4 = (1-x)(1-x^2)^4$.

Step 1: The x^4 coefficient of the even-power factor is $\binom{4}{2} = 6$.

Step 2: The $-x$ factor has no x^3 term to multiply, so contributes zero.

Worked derivation:

$$[x^4](1-x)(1-x^2)^4 = [x^4](1-x^2)^4 = \binom{4}{2} = 6.$$

Why: Combining conjugate factors exposes parity and prevents signed-term bookkeeping errors. The unmatched $(1-x)$ affects odd powers only at this degree. The x^4 term is drawn from the second-order choice of four factors of $-x^2$, which has positive sign despite the outer minus factor.

Why not the others: (A) reverses the sign of an even selection; (B) treats the coefficient as a single signed term; (C) counts one choice instead of the six ways to select two x^2 factors.

Key point: Simplify before extracting a signed coefficient.

Q41 / A Equal-length GP blocks and an infinite geometric sum

Answer: (A) 8.

Given: Let the first term be $a > 0$, with $0 < r < 1$.

Formula/Principle: Shifting a three-term block three places multiplies its sum by r^3.

Step 1:

$$r^3 = \frac{7/8}{7} = \frac{1}{8} \;\Rightarrow\; r = \frac{1}{2}.$$

Step 2:

$$a(1+r+r^2) = 7 \;\Rightarrow\; \frac{7}{4}a = 7 \;\Rightarrow\; a = 4.$$

Step 3:

$$S_\infty = \frac{a}{1-r} = \frac{4}{1/2} = 8.$$

Worked derivation:

$$S_\infty = \frac{7}{1-r^3} = \frac{7}{1-1/8} = 8.$$

Why: Consecutive blocks of equal length themselves form a geometric progression, so the given block ratio determines the underlying common ratio. Positivity and decrease ensure convergence and exclude irrelevant negative roots.

Why not the others: (B) omits all later blocks; (C) doubles the first block rather than summing its geometric continuation; (D) uses an incorrect common ratio.

Key point: Compare equal-length blocks before recovering individual terms.

Q42 / C common-root condition

Answer: (C) 2.

Given: The first polynomial has roots 1 and 2.

Formula/Principle: Test each possible common root in the parameterised second polynomial, then combine distinct parameter solutions.

Step 1: At x=1, the second polynomial is $a^2-a-1=0$, giving $a=(1\pm\sqrt{5})/2$.

Step 2: At x=2, it is $(a-1)^2=0$, giving a=1.

Step 3: The three values are distinct; the conjugate pair sums to 1, so total 2.

Worked derivation:

$$p(x)=(x-1)(x-2),\quad q(1)=a^2-a-1=0,\quad q(2)=(a-1)^2=0;\quad \sum a=\frac{1+\sqrt{5}}{2}+\frac{1-\sqrt{5}}{2}+1=2.$$

Why: A common root must be either 1 or 2, and both branches can generate parameters. The repeated factor in q(2) gives one distinct parameter, not two copies of a=1; the question explicitly sums distinct real values.

Why not the others: (A) keeps only the golden-ratio pair; (B) double-counts a=1; (D) keeps just one root of the first branch.

Key point: Branch over every candidate common root and deduplicate parameters.

Q43 / C standard-deviation constraint

Answer: (C) 6.

Given: Mean fixes c through $17=2(5)+c$.

Formula/Principle: Translation leaves SD unchanged; scaling by 2 doubles it.

Step 1: c=7.

Step 2: $y-\bar{y}=2(x-\bar{x})$, hence SD 2·3=6.

Worked derivation:

$$c=17-10=7,\quad \sigma_y=\sqrt{E[(2(x-\bar{x}))^2]}=2\sigma_x=6.$$

Why: The offset determines the centre but cancels from every deviation. Variance scales by four, while standard deviation scales by two. The square root in standard deviation is why doubling all deviations doubles SD rather than multiplying it by four.

Why not the others: (A) ignores scaling; (B) adds the shift; (D) uses the variance factor for SD.

Key point: Separate centre translation from spread scaling.

Q44 / D separable initial-value problem

Answer: (D) x=1.

Given: Initial-value problem and positive branch.

Formula/Principle: Separate $y^{-2}dy=x,dx$.

Step 1: $-1/y=x^2/2-1$, so $y=1/(1-x^2/2)$.

Step 2: Setting y=2 gives $x^2=1$ and positive x=1.

Worked derivation:

$$-\frac{1}{y}=\frac{x^2}{2}-1;\quad \frac{1}{2}=1-\frac{x^2}{2}\Rightarrow x=1.$$

Why: The solution is positive before its singularity at $x=\sqrt{2}$; the other algebraic root is negative and does not answer the question. The initial condition determines the integration constant; without it the differential equation admits a family of curves and no unique crossing time.

Why not the others: (A) gives y=8/7; (C) is singular; (B) lies beyond this branch.

Key point: Check domain after solving a separable IVP.

Q45 / B one-sided absolute-value limit

Answer: (B) (−1/2,1/2).

Given: Denominator factors as (x−1)(x+1).

Formula/Principle: |x−1|=−(x−1) on the left and x−1 on the right.

Step 1: Left expression is −1/(x+1)→−1/2.

Step 2: Right expression is +1/(x+1)→+1/2.

Worked derivation:

$$f(x)=\begin{cases}-1/(x+1) & x<1,\\ 1/(x+1) & x>1;\end{cases}\qquad (L_-,L_+)=(-\tfrac{1}{2},\tfrac{1}{2}).$$

Why: Cancelling before resolving the absolute value loses the sign change. Both one-sided limits exist, but the two-sided limit does not. The finite one-sided values do not match, so continuity cannot be restored by assigning any single value to f at x=1.

Why not the others: (A) ignores the left sign; (C) swaps order; (D) ignores the right sign.

Key point: Split absolute-value expressions by side.

Physics

Q46 / C significant-digit inference

Answer: (C) 1.50%.

Given: Two independent worst-case bounds.

Formula/Principle: Relative product bounds add to first order.

Step 1: Length fraction 0.02/2.40=0.00833; width 0.01/1.50=0.00667.

Step 2: Sum 0.0150=1.50%.

Worked derivation:

$$\frac{\delta A}{A}\simeq\frac{\delta l}{l}+\frac{\delta w}{w}=\frac{0.02}{2.40}+\frac{0.01}{1.50}=0.0150.$$

Why: Maximum bounds may reinforce; root-sum-square applies to a different statistical interpretation. Second-order terms are small here. Because the uncertainty asks for a bound, the sign of each measurement error can be chosen to maximize area change; this differs from propagation of independent standard deviations.

Why not the others: (A) keeps width alone; (B) length alone; (D) overrounds.

Key point: Add relative worst-case errors for products.

Q47 / C constraint acceleration

Answer: (C) 2.4 m s⁻².

Given: The accelerating support frame has downward effective gravity 12 m s⁻².

Formula/Principle: Relative Atwood acceleration $(m_2-m_1)g_{eff}/(m_1+m_2)$.

Step 1: Mass ratio is 1/5.

Step 2: Multiply by 12 to obtain 12/5=2.4 m s⁻².

Worked derivation:

$$g_{eff}=10+2=12,\quad a_{rel}=\frac{3-2}{3+2}(12)=2.4\,\mathrm{m,s^{-2}}.$$

Why: The upward support acceleration adds downward pseudo-force to both masses. Laboratory acceleration would require transforming frames, but the question asks relative motion. The heavier mass still moves down relative to the pulley even though its laboratory acceleration can be altered by the elevator's motion.

Why not the others: (A) misses frame effect; (B) confuses support motion with relative acceleration; (D) uses the wrong mass factor.

Key point: State the frame before force balance.

Q48 / D elastic-energy density

Answer: (D) Factor 4.

Given: Material modulus and strain remain constant.

Formula/Principle: Elastic energy density $u = Y\epsilon^2/2$; total energy uAL.

Step 1: Energy density is unchanged.

Step 2: Volume becomes (2A)(2L)=4AL.

Worked derivation:

$$U = \tfrac{1}{2}Y\epsilon^2 AL, \quad U' = \tfrac{1}{2}Y\epsilon^2(2A)(2L) = 4U.$$

Why: Fixing absolute extension would change strain and the answer; the stated fixed strain makes stored energy proportional to volume. Young's modulus and strain set the local energy density; changing both geometric dimensions affects only the amount of material under this constraint.

Why not the others: (A) ignores extra volume; (B) doubles one dimension; (C) adds a nonexistent strain factor.

Key point: Identify the held-fixed elastic quantity.

Q49 / B conduction thermal resistance

Answer: (B) 1/2.

Given: Same heat current through both layers.

Formula/Principle: Thermal resistance $R = L/(kA)$; temperature drop follows resistance.

Step 1: $R_1 = 2d/(2kA) = d/(kA)$.

Step 2: $R_2 = d/(kA) = R_1$; fraction 1/2.

Worked derivation:

$$\frac{\Delta T_1}{\Delta T_{total}} = \frac{R_1}{R_1 + R_2} = \frac{d/(kA)}{2d/(kA)} = \frac{1}{2}.$$

Why: Double thickness is exactly offset by double conductivity. Comparing thickness alone ignores the second material property. Series flow has no alternate heat path, so equal thermal resistances must take equal shares of the imposed end-to-end temperature difference.

Why not the others: (A) treats conductivity as the only changing resistance factor; (C) compares thicknesses 2d:d while ignoring conductivities; (D) assigns an unsupported three-to-one resistance ratio.

Key point: Compare complete thermal resistances.

Q50 / D mean free path trend

Answer: (D) Path halves; speed unchanged.

Worked derivation:

$$\lambda = \frac{k_B T}{\sqrt{2}\pi d^2 p} \Rightarrow \lambda' = \lambda/2, \qquad \bar{v} \propto \sqrt{T} \Rightarrow \bar{v}' = \bar{v}.$$

Why: Higher pressure at fixed T means greater number density and more frequent collisions per distance. Thermal molecular speed depends on T and mass, neither changed. A higher number density shortens the typical travel before collision even though the speed distribution remains the same at equal temperature.

Why not the others: (B) changes speed without T change; (C) reverses dependencies; (A) ignores denser collisions.

Key point: Separate collision length from thermal speed.

Q51 / B velocity-acceleration relation

Answer: (B) √15 m s⁻¹.

Given: a varies with position.

Formula/Principle: Chain rule $a = v, dv/dx$.

Step 1: Integrate $v, dv = 2x, dx$ between stated states.

Step 2: $(v^2 - 9)/2 = 4 - 1 = 3$, hence v²=15.

Worked derivation:

$$\int_3^v v, dv = \int_1^2 2x, dx \Rightarrow \frac{v^2 - 9}{2} = 3 \Rightarrow v = \sqrt{15}\,\mathrm{m, s^{-1}}.$$

Why: A constant-acceleration formula is inapplicable. The +x direction selects the positive root. The integral accumulates the work associated with each position; using the acceleration at x=2 throughout would exaggerate the speed increase.

Why not the others: (A) omits work; (C) assumes constant a; (D) adds velocities linearly.

Key point: Use $v, dv/dx$ for position-dependent acceleration.

Q52 / A reflection phase at boundary

Answer: (A) Approximately fixed end.

Worked derivation:

$$Z = \sqrt{T\mu}, \quad Z_2 \gg Z_1 \Rightarrow r_{displacement} = \frac{Z_1 - Z_2}{Z_1 + Z_2} < 0.$$

Why: The high-impedance string resists displacement at the junction, giving a negative reflection coefficient. Some energy may transmit, so the fixed-end analogy describes the limit, not an absolute wall.

Why not the others: (B) gives upright reflection; (C) frequency does not change sign; (D) speed in the original string stays fixed.

Key point: Reflection phase depends on impedance contrast.

Q53 / D field cancellation locus

Answer: (D) x=a.

Given: The positive-charge fields oppose between the charges.

Formula/Principle: Equalise inverse-square field magnitudes.

Step 1: $q/x^2 = 4q/(3a - x)^2$.

Step 2: Positive distances yield 3a−x=2x, hence x=a.

Worked derivation:

$$\frac{kq}{x^2} = \frac{4kq}{(3a - x)^2} \Rightarrow 3a - x = 2x \Rightarrow x = a.$$

Why: The cancellation lies nearer the weaker charge. Outside the interval both field vectors point the same way and cannot cancel. The root x=a satisfies both the distance inequality 0<x<3a and the qualitative expectation that the stronger charge's field balances farther away.

Why not the others: (B) assumes equal charges; (C) is nearer the stronger charge; (A) ignores opposed fields between them.

Key point: Decide vector directions before equating magnitudes.

Q54 / C bridge balance with temperature

Answer: (C) 120 Ω.

Given: P/Q remains 1/2.

Formula/Principle: Bridge balance $P/Q = R/S$.

Step 1: Require 60/S=100/200.

Step 2: S=120 Ω.

Worked derivation:

$$\frac{100}{200} = \frac{60}{S} \Rightarrow S = 120\,\Omega.$$

Why: Increasing R requires proportional increase in S to preserve the ratio. Ideal balance does not depend on supply voltage. The warming affects only R in the stated bridge; changing the complementary arm S by the same factor restores the galvanometer's null condition.

Why not the others: (A) gives ratio one; (B) leaves imbalance; (D) gives ratio one quarter.

Key point: Preserve the arm ratio under temperature change.

Q55 / C rotating rod motional emf

Answer: (C) 3/4.

Given: Local speed is ωr.

Formula/Principle: $d\mathcal{E} = B\omega r, dr$.

Step 1: Integrate r from L/2 to L.

Step 2: Divide by integral 0 to L.

Worked derivation:

$$\frac{\mathcal{E}_{outer}}{\mathcal{E}_{full}} = \frac{L^2 - (L/2)^2}{L^2} = \frac{3}{4}.$$

Why: The outer half has greater speed, so contributes more than half the emf. The comparison cancels B and ω. This outer annulus-like segment contributes disproportionately because the local linear velocity is proportional to distance from the pivot.

Why not the others: (A) is the inner half; (B) treats speed as uniform; (D) includes the missing inner part.

Key point: Integrate local motional emf along radius.

Q56 / B inductor-capacitor voltage cancellation

Answer: (B) √10 A.

Given: Net reactance 10 Ω.

Formula/Principle: $Z = \sqrt{R^2 + (X_L - X_C)^2}$.

Step 1: Z=√(900+100)=10√10 Ω.

Step 2: I=100/(10√10)=√10 A.

Worked derivation:

$$Z = \sqrt{30^2 + (40-30)^2} = 10\sqrt{10}\,\Omega, \quad I = \frac{100}{Z} = \sqrt{10}\,\text{A}.$$

Why: Inductive and capacitive voltages oppose in phase, so reactances subtract before quadrature with resistance. The remaining 10-ohm reactance is not zero, so the circuit is not at resonance and the rms current is below the purely resistive value.

Why not the others: (A) adds impedances incorrectly; (C) ignores residual reactance; (D) underestimates Z.

Key point: Account for phase in impedance.

Q57 / D real-virtual image sign

Answer: (D) v negative, magnification positive.

Worked derivation:

$$f > 0,\ -f < u < 0; \quad 1/v = 1/f + 1/u < 0, \quad m = v/u > 0.$$

Why: The image is on the object's side of the lens; both object and image distances are negative. Their ratio is positive, matching upright orientation.

Why not the others: (B) places the virtual image on transmitted side; (C) predicts inversion; (A) ignores distance dependence.

Key point: Apply one sign convention consistently.

Q58 / B diffraction central width

Answer: (B) 6 mm.

Given: Initial full width is 12 mm.

Formula/Principle: First-minimum displacement $y_1 \simeq D\lambda/a$.

Step 1: Double a, so each minimum moves to ±3 mm.

Step 2: Full span is 6 mm.

Worked derivation:

$$W_0 = 2(6) = 12\,\text{mm}, \quad W' = \frac{2D\lambda}{2a} = W_0/2 = 6\,\text{mm}.$$

Why: The marked ± values are half-width positions. A wider aperture narrows diffraction, and the span must count both sides. The position of each minimum scales inversely with slit width, while the central maximum spans the two first-minimum positions.

Why not the others: (A) is new half-width; (C) ignores change; (D) reverses scaling.

Key point: Separate fringe position from full width.

Q59 / A threshold frequency and work function

Answer: (A) 2.10 eV and about 590 nm.

Given: Photon energy is 3.10 eV, stopping energy 1.0 eV.

Formula/Principle: $eV_s = hc/\lambda - \phi$, $\lambda_0 = hc/\phi$.

Step 1: φ=3.10−1.0=2.10 eV.

Step 2: λ0=1240/2.10≈590 nm, shorter than non-emitting 600 nm.

Worked derivation:

$$\phi = 1240/400 - 1 = 2.10\,\text{eV}, \quad \lambda_0 = 1240/2.10 \approx 590\,\text{nm}.$$

Why: Stopping voltage measures kinetic energy, not the entire photon energy. The threshold agrees with both observations. A threshold of about 590 nm predicts emission for the 400 nm test and no emission for 600 nm, checking both experimental observations.

Why not the others: (B) ignores kinetic energy; (C) mistakes stopping energy for φ; (D) adds instead of subtracting.

Key point: Cross-check threshold against emission/no-emission trials.

Q60 / A p-n junction bias

Answer: (A) X forward conducts, Y reverse leaks weakly.

Worked derivation:

$$V_p - V_n > 0 \Rightarrow \text{forward bias}, \qquad V_p - V_n < 0 \Rightarrow \text{reverse bias}.$$

Why: Forward bias lowers the junction barrier and injects majority carriers; reverse bias widens depletion and limits current before breakdown. Equal resistors do not erase rectification.

Why not the others: (B) ignores polarity; (C) reverses the definition; (D) ignores forward conduction.

Key point: Determine bias from p-to-n voltage sign.